Components Degradation and Classification - Guide

This document provides a detailed overview of the files found in the 'components degradation and classification' directory, explaining the purpose of each document and detailing the fields of the Excel columns.

1. Anomalies and Defects (Lista Anomalie)

File: 130208_ListaAnomalieOriginale.xlsm

This macro-enabled Excel file serves as the main database for tracking component degradation and anomalies. It connects the Work Breakdown Structure (WBS) with specific defects.

Sheet: ListaAnomalieCompleta - Contains the comprehensive mapping of files, WBS codes, and the specific anomalies.

Columns:

• #: An incremental ID or index.
• **NOME FILE (File Name)**: The reference file name associated with the element (e.g., 01_StrutturaDiFondazione.xlsx).
• **CODICE WBS (WBS Code)**: The Work Breakdown Structure code identifying the architectural/structural element (e.g., S.F.01.01.01.01).
• **GRAVITÀ ANOMALIA (Anomaly Severity)**: Defines how critical the defect is, graded as M (Minore / Minor), S (Seria / Serious), or G (Grave / Severe).
• **NOME ANOMALIA (Anomaly Name)**: The name of the defect (e.g., 'Bolle d'aria' / Air bubbles).
• **DESCRIZIONE ANOMALIA (Anomaly Description)**: A detailed text explaining the defect's characteristics.
• **PARAMETRO DI MISURA (Measurement Parameter)**: The metric used to evaluate the anomaly (e.g., size and depth of the holes).
• **COMPORTAMENTO (Behavior)**: The structural or material behavior resulting from the anomaly.
• **NOME FOGLIO (Sheet Name)**: The specific Excel sheet referencing this component.
• **CODICE ANOMALIA (Anomaly Code)**: An alphanumeric code (e.g., A40) for uniquely identifying the defect.
• **RSL COMPONENTE (Reference Service Life)**: The expected lifespan of the component in years.

File: 120914_ListaAnomalieCompleta.xlsx

An older, raw version of the anomalies list. It contains the data directly without headers in the first row, mapping similar fields as the macro-enabled version.

2. Work Breakdown Structure (WBS)

Files: 130221_WBS.xlsx and 130221_WBS.pdf

These documents describe the hierarchical classification of building elements according to standard UNI conventions (Classes of Technological Units down to Functional Elements).

Sheet: WBS

Columns:

• #: Index.
• **CLASSE DI UNITÁ TECNOLOGICHE (Class of Technological Units)**: Broadest category (e.g., Struttura / Structure).

• **COD.**: The code for the Class (e.g., S).
• **UNITÁ TECNOLOGICHE (Technological Units)**: Sub-category (e.g., Struttura di fondazione / Foundation).
• **COD.**: The code for the Unit (e.g., S.F).
• **CLASSE DI ELEMENTI TECNICI (Class of Technical Elements)**: Finer categorization (e.g., Strutture di fondazione dirette).
• **COD.**: The code for the Technical Element Class (e.g., S.F.01).
• **ELEMENTI TECNICI (Technical Elements)**: The specific element description (e.g., Plinti, travi rovesce, piastra).
• **COD.**: The final WBS code for the specific element (e.g., S.F.01.01.01.01).
• **ELEMENTI/STRATI FUNZIONALI (Functional Elements/Layers)**: The material or functional layer (e.g., Calcestruzzo armato / Reinforced concrete).

3. Reference Service Life (RSL)

Files: 130416_ListaRSL.xlsm and RSL.pdf

These documents map the WBS elements to their Reference Service Life (RSL), indicating the expected lifespan of each structural component before major maintenance or failure.

Sheet: RSL

Columns:

• **NOME FILE (File Name)**: The reference file name for the component.
• **CODICE WBS (WBS Code)**: The Work Breakdown Structure code (e.g., S.F.01.01.01.01).
• **NOME FOGLIO (Sheet Name)**: The specific element name.
• **RSL COMPONENTE [anni] (Component RSL [years])**: The estimated lifespan of the component in years (e.g., 45 years).